

PPE detection with YOLOv8: from "is there PPE?" to "who is non-compliant?"

Problem

Safety staff check PPE in site photos by eye, which is slow and inconsistent. A detector that pre-screens photos lets the reviewer focus on doubtful cases.

Success criteria (set before evaluation)

- Recall ≥ 0.85 per class (a missed PPE item is the costly error)
- mAP50 ≥ 0.80 on the 20 % validation split
- Correct on new, unseen single-worker photos

Deliverable: one public GitHub repository, reproducible in Google Colab with no credentials.

github.com/arqmanu/M4U3_ppe-detection-yolov8

This model is an assistive tool for preliminary screening only. It produces false negatives and false positives. It must not be used as the sole verifier for life-safety decisions.

Two PPE classes, with one deliberate rule: an open vest is not valid PPE

CLASS	LABEL	DO NOT LABEL
head_protection	Company dark-blue protective cap; safety helmet (incl. white)	Ordinary caps
high_visibility_clothing	Closed yellow/orange vest with reflective stripes; yellow-grey reflective fleece; closed red visitor vest	Open vests, red company T-shirts, clothing without stripes

The open-vest rule makes the model separate *wearing a garment* from *wearing it correctly*. It is an academic project decision, not a legal rule.

Box rules: one box per visible item, tight to the item; partial PPE only if it can be identified with confidence. Contract: docs/class_definitions.md



Ground truth: cap (green), closed vest (orange)



Back view, same rules

134 photos of my own workplace, labelled in Roboflow and frozen for keyless reproduction

134

images, consent
signed, faces
obscured

107/27

train / validation (80 / 20)

308

labelled PPE objects

- **Labelling:** Roboflow Astra Auto Label proposals, then a full manual review
- **Preprocessing:** auto-orient, fit within 640×640 ; no augmentation
- **Model:** YOLOv8n from `yolov8n.pt`, 30 epochs, imgsz 640, batch 16, seed 0, Ultralytics 8.2.103
- **5 new test photos** never uploaded to Roboflow. One was replaced after I found it in the validation split

Run all, with no keys

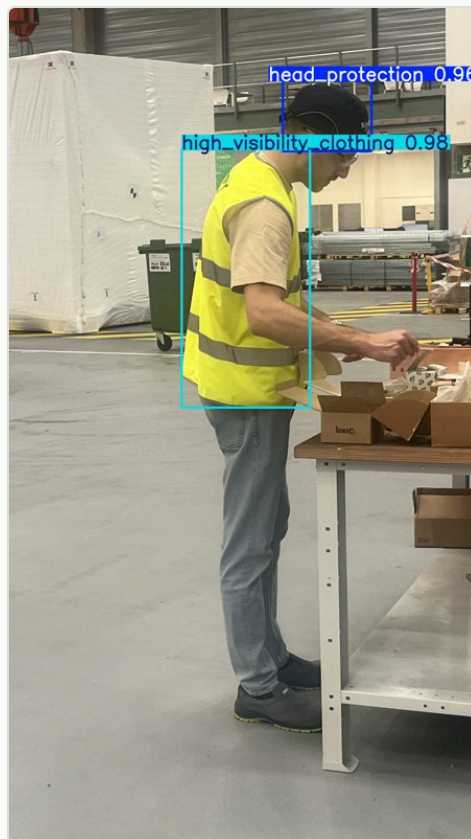
1. Roboflow **v3** export → GitHub Release `v1.0`
2. Notebook downloads it and checks the **SHA-256**
3. GPU → retrain · no GPU → load published `best.pt` (SHA-256 checked)
4. Metrics, evidence and error table saved automatically

Proof: fresh-runtime Run all on Colab CPU, 25 Sep 2026, 1.6 min, mAP50 0.922 (training run: 0.923).

A generic model sees "person". The custom model sees PPE, with mAP50 0.92



Baseline COCO yolov8n.pt :
"person 0.51", no PPE classes

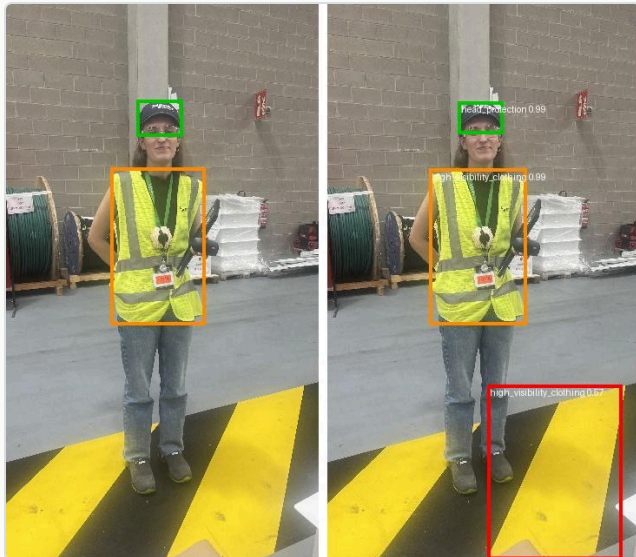


Custom model on a new photo:
cap 0.96, vest 0.98

VALIDATION (27 IMGS)	P	R	MAP50	MAP50-95
all	0.833	0.897	0.923	0.717
head_protection	0.920	0.860	0.898	0.646
high_vis_clothing	0.746	0.933	0.947	0.789

- **met** recall ≥ 0.85 per class, mAP50 ≥ 0.80
- Clothing is found reliably, but **over-detected** (precision 0.75)
- Caps are precise, but **missed more often** when small or distant
- **not met** reliable on crowded or distant scenes

Errors come from colour patterns, small objects and an unlearned rule



FP: floor hazard stripes seen as a vest (0.57)



FN: distant caps missed (×2)



FP: my open-vest test photo, accepted as valid (0.52)

3 false positives · 3 false negatives

- **FP:** floor stripes · red T-shirt · shadow next to a head
- **FN:** distant heads · crowded scene · cap seen from above
- **Open-vest rule not learned:** colour and stripes dominate the "is it closed?" cue · 1 label error found

Next data, prioritised

1. Fix the label contract and relabel (done in iteration 2)
2. Add small, distant and crowded workers
3. Add hard negatives: floor markings, red shirts, shadows

"No helmet detected" ≠ "no helmet worn". So the violation became a class

Added `no_head_protection` (head box) and `no_high_visibility_clothing` (torso box; **open vests included**). Same images, same split and same recipe (Roboflow v5 · Release v2.0).

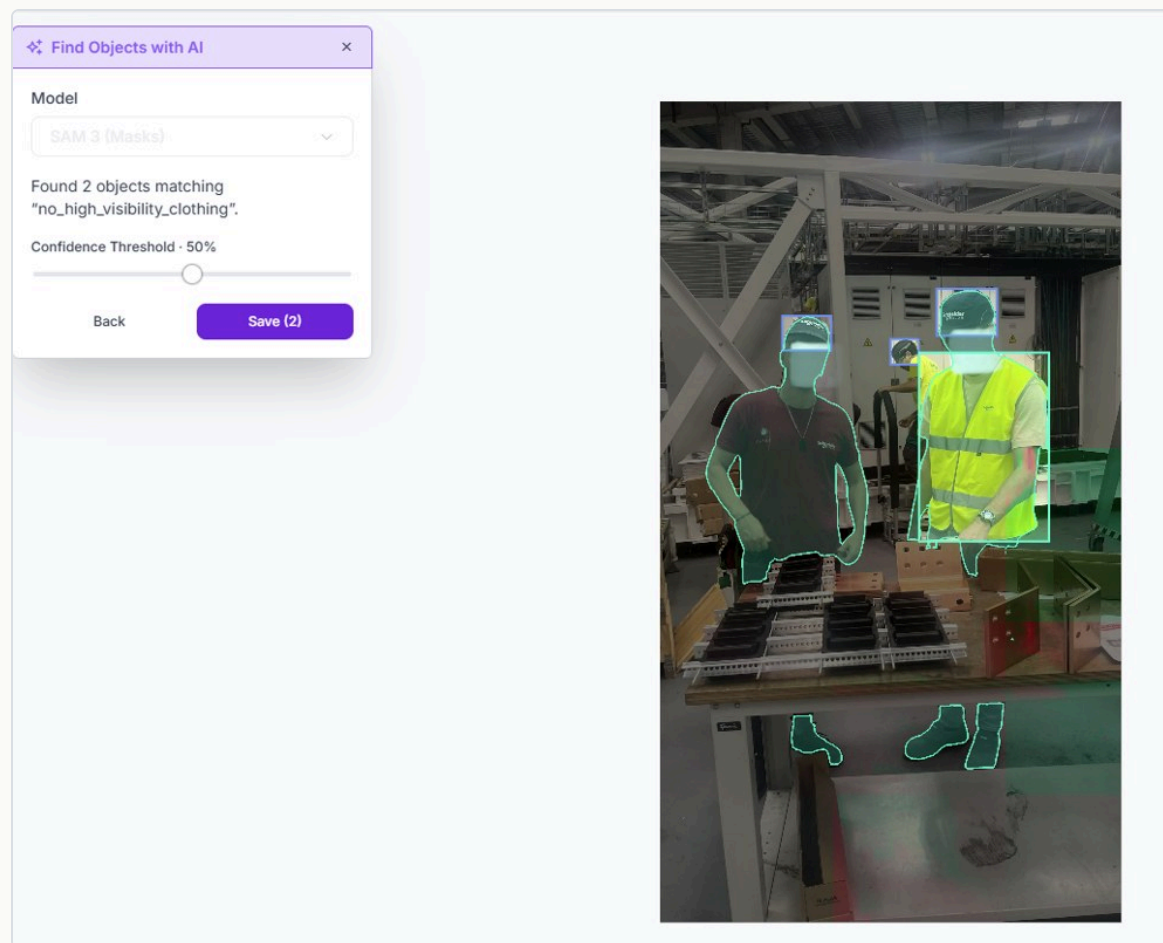
	ITERATION 1	ITERATION 2
mAP50 / Recall	0.923 / 0.897	0.905 / 0.782
<code>head_protection</code> R	0.860	0.713
<code>no_high_vis</code> P / R	n/a	0.711 / 0.875
<code>no_head</code> examples	n/a	23 total, 3 in val

Result: it answers the right question, but violations are rare in a compliant workplace. **Iteration 1 remains the better PPE detector; iteration 2 is a proof of concept.** Next step: stage violation photos to balance the classes.



Same new photo. Left: iteration 1 (cap only). Right: iteration 2 flags the missing vest (`no_high_visibility_clothing` 0.66)

SAM 3 segments people very well, but it cannot express "without a vest"



Prompt: "upper body of a worker wearing a red t-shirt, without a reflective safety vest"

What helped

- Precise masks from a text prompt, with no clicks and no training
- Useful for "find this *visible* thing" classes

What failed

- It **also selected the worker wearing a vest**: absence cannot be prompted
- Shoes returned as separate fragments: the mask-to-box "**geometry tax**" would stretch the box to the feet
- It cannot know project rules (open vest = invalid)

Decision: suggestions not saved; violation boxes drawn by hand against a written contract.

A screening aid with a human in the loop, not an enforcement tool

Privacy and data

- My own workplace photos; written consent covers publication
- Faces obscured in most images; the model detects objects, not people
- No identity, tracking or biometric data; images ≤ 640 px

Risk

- **False negatives** (missed PPE or a missed violation) are the costly error, so thresholds should favour recall
- False positives cost reviewer time, and can hide a violation

Do not use for

- Automated enforcement, discipline or surveillance
- Crowded or distant scenes, other sites, other PPE types
- Proving absence of PPE with iteration 1

Licensing

- Code: MIT · Dataset: CC BY 4.0 (own photos) · Ultralytics: AGPL-3.0

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